

## Two Measurements per Technician

AP Statistics · Unit 4 · Topic 4.4 · B: 2027-02-01 · E: 2027-02-03 · TEACHER KEY

### CED TETHER

- **Skill 1.E** — Identify an appropriate inference method for significance tests.
- **Skill 1.F** — Identify null and alternative hypotheses.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **EU VAR-7** — The t-distribution may be used to model variation.
- **LO VAR-7.B** — Identify an appropriate testing method for a population mean with unknown  $\sigma$ , including the mean difference between values in matched pairs. [Skill 1.E]
- **EK VAR-7.B.1** — The appropriate test for a population mean with unknown  $\sigma$  is a one-sample t-test for a population mean.
- **EK VAR-7.B.2** — Matched pairs can be thought of as one sample of pairs. Once differences between pairs of values are found, inference for significance testing proceeds as for a population mean.
- **LO VAR-7.C** — Identify the null and alternative hypotheses for a population mean with unknown  $\sigma$ , including the mean difference between values in matched pairs. [Skill 1.F]
- **EK VAR-7.C.1** — The null hypothesis for a one-sample t-test for a population mean is  $H_0: \mu = \mu_0$ , where  $\mu_0$  is the hypothesized value. Depending upon the situation, the alternative hypothesis is  $H_a: \mu < \mu_0$ , or  $H_a: \mu > \mu_0$ , or  $H_a: \mu \neq \mu_0$ .
- **EK VAR-7.C.2** — When finding the mean difference,  $\mu_D$ , between values in a matched pair, it is important to define the order of subtraction.
- **LO VAR-7.D** — Verify the conditions for the test for a population mean, including the mean difference between values in matched pairs. [Skill 4.C]
- **EK VAR-7.D.1** — In order to make statistical inferences when testing a population mean, we must check for independence and that the sampling distribution is approximately normal: a. To check for independence: (i) Data should be collected using a random sample or a randomized experiment. (ii) When sampling without replacement, check that  $n \leq 10\% N$ .
- b. To check that the sampling distribution of  $\bar{x}$  is approximately normal (shape): (i) If the observed distribution is skewed,  $n$  should be greater than 30. (ii) If the sample size is less than 30, the distribution of the sample data should be free from strong skewness and outliers.

**Essential question:** How does the study structure determine the parameter, hypotheses, and conditions for a mean test?

**DOK-3 skill:** 4.C · **FRQ pattern:** paired-test-setup-and-consequences

**DOK rationale:** Part (c) coordinates repeated measures, subtraction order, tail direction, independent sample size, and shape evidence, then traces the uncertainty and research-question consequences of the rejected setup.

**First take** → rules + (a)–(c) → turn in

**DOK:** 1 → 3

**Minutes:** 45

**Teacher does**

- Display the board and ask for one sentence using the study description.
- At minute 5, read the rules box aloud once; students start (a).
- Return to the board at minute 33. Students finish the shortened parts and justify their judgment.
- Collect at the door. Score part (c) only using the three required elements.

### Students do

- Write a one-sentence first take before any discussion.
- Work (a) and (b) from the rules box and the stem.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

### Questions to ask

- Which specific number or study detail supports your choice?
- What claim would go beyond the evidence, and why?

### Adult role

Circulate and ask for evidence. Accept a concise response; do not require omitted calculations or a second full inference write-up.

### [WARN] LOOK FOR

- Choosing a speaker without a reason.
- Copying numbers without explaining what they support.

## SCORING PART (c) — E / P / I

### Expected elements

- **paired-units** (required) — Justifies a one-sample test of the 18 within-technician differences
- **correct-hypotheses** (required) — Uses the population mean difference, zero null, and lower-tail alternative
- **rejects-independent-times** (required) — Explains that the two times from each person are linked

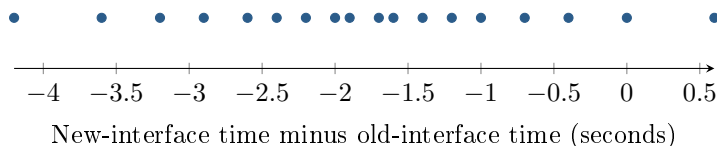
|          |                                                                                                                                                                       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>E</b> | Justifies the paired units and explains the lower-tail population hypotheses using new minus old and the reduction question.                                          |
| <b>P</b> | Shows substantive valid reasoning for at least one required element, but does not meet all E requirements. Credit correct reasoning even when another claim is wrong. |
| <b>I</b> | Shows no substantive valid reasoning for any required element; a name, unsupported choice, or copied number alone does not count.                                     |

## [STAR] TODAY'S PROBLEM — annotated

Suppose a company takes an SRS of 18 of its 300 technicians. Each completes the same task with old and new interfaces; interface order is randomly assigned. The plot shows  $d = \text{new time} - \text{old time}$ . Researchers ask whether the new interface reduces the population mean time.

**Nia:** “The technician seems to be the independent unit, so I would build one difference per person; the subtraction order matters for the tail.”

**Colin:** “There are 36 recorded times, so I would use two independent samples of 18 and a two-sided alternative.”



### FIRST TAKE (5 min)

Would you compare each technician's two times or treat them as separate people? Explain in one sentence.

**Key:** Ungraded. Everyday language is enough before the discussion; formal vocabulary belongs in the finish.

*Rules callout “SET UP A TEST FOR A MEAN (keep on the page)” is printed on the student sheet.*

**DOK 1** (a) Define  $\mu_d$  in context.

**Key:**  $\mu_d$  is the mean new-minus-old task time for all 300 technicians.

**DOK 2** (b) Check the random-sample, 10 percent, and small-sample shape conditions in three short notes.

**Key:** The 18 technicians were randomly sampled.  $18 \leq 0.10(300) = 30$ . Since  $n = 18 < 30$ , inspect the differences: the plot shows no strong skewness or outliers.

**DOK 3** (c) In 3–4 sentences, choose a setup and justify it. State the hypotheses using the given subtraction order and reduction question, then explain why the 36 times are not 36 independent people.

**Key:** Nia is justified: use a matched-pairs one-sample  $t$ -test on the 18 differences. For new minus old, a reduction is negative, so  $H_0 : \mu_d = 0$  and  $H_a : \mu_d < 0$ . Each technician supplies two linked times, so there are 18 independent people, not 36. Treating the times as independent loses the pairing, and a two-sided alternative asks about any change rather than the planned reduction.

### EXIT

One thing the rules box changed about my first take: \_\_\_\_\_